

# BioDecisionBench: A Leak-Proof Typed-Decision Benchmark for Biological Sequences, and a Seed-Averaged Comparison of Decision Architectures

Anonymous Authors  
(working draft — venue unbound)

October 2026

## Abstract

Can one model answer “is this promoter active?” (a yes/no judgement), “which fold does this protein adopt?” (a choice among labels), and “how fit is this mutant?” (an ordinal score) through a single decision interface? Typed-decision interfaces — which encode the question and the candidate answers as text and let one *shared scorer* rank candidates, with no per-task output heads — are being adopted for exactly this setting, but have never been compared against conventional task heads under matched compute. We release **BioDecisionBench**: 290 biological sequence decision tasks spanning DNA, protein and RNA, unified under four typed decision primitives — Noul (yes/no judgement), Choice (mutually exclusive labels), Score (ordinal value), and multi-Noul (multi-label) — with a leak-proof admission protocol that keeps legacy blind-test sets verbatim, quarantines 1,984 conflicting-label groups instead of majority-voting them, and verifies zero within-task train/test sequence overlap. On this benchmark we run, to our knowledge, the first seed-averaged three-way comparison of decision architectures: a shared typed-decision scorer, per-task heads on the same encoder, and a frozen small language model scored by full-candidate likelihood. Averaged over three seeds, per-task heads match or beat the shared scorer — Noul AUROC  $0.629 \pm 0.050$  vs.  $0.478 \pm 0.051$  (disjoint SEMs), Choice and Score at parity — while the shared scorer is markedly less stable (run-std 0.087 vs 0.030). The shared scorer’s claimed advantages survive only qualitatively: its zero-shot transfer is at the level of the untrained base, its cross-family transfer is weak ( $+0.036 \pm 0.034$ , classification only), and a fresh task head catches its zero-shot performance with  $\approx 32$  labels. Single-run differences of  $\pm 0.05$ – $0.1$  reverse architectural conclusions in this regime; we document a complete case in which a single-run verdict is reversed by seed averaging, and propose  $\geq 3$ -seed averaging with run-std reporting as the minimum bar for such comparisons.

## 1 Introduction

A single biological sequence model should ideally answer qualitatively different questions through one interface: “is this DNA segment a promoter?” (a yes/no judgement), “which superfamily does this protein belong to?” (a choice among labels), “how fit is this sequence variant?” (an ordinal score), and “where is this protein localized?” (a multi-label assignment). *Typed-decision* interfaces serve exactly this purpose: the question and the candidate answers are rendered as text, encoded together with the sequence, and a single *shared decision scorer* outputs a probability per candidate — no task-specific output heads, no free-text generation. This design is being adopted in practice, by commercial systems (Almeida, 2026) and by open reimplementations (Stepanov et al., 2025; TianyuCodings, 2026; TheoLeeCJ, 2025), on the promise of one small model, dynamic candidate sets, and zero-adaptation-cost new tasks. Yet two basic questions have never been answered under controlled conditions. **Q1**: does a shared typed-decision scorer match or beat conventional per-task output heads at matched compute? **Q2**: does the “dynamic candidates / zero adaptation” selling point translate into a measurable performance or cost advantage?

Answering these questions runs into three obstacles. First, biological sequence evaluation is fragmented: GUE (Zhou et al., 2024), Genomic Benchmarks (Gresova et al., 2022), TAPE (Rao et al., 2019), ProteinGym (Notin et al., 2023) and friends each define their own task formats, splits and metrics, so decision architectures cannot be compared in one protocol. Second, leakage auditing is largely absent: several upstream “train” pools on public hubs *contain the blind-test members of prior studies*, so a random re-split silently fabricates a new test set. Third

— the methodological spine of this paper — *evaluation variance in this regime has never been quantified*. In our small-budget multi-task setting, two runs of the same configuration and seed differ by  $\pm 0.05$ – $0.1$  on task-macro metrics, enough to reverse the winner of an architecture comparison (Section 5.1). Any single-run architectural conclusion is therefore unreliable, including our own early one.

**Approach.** We build a benchmark first and a measurement protocol second, then compare architectures. BioDecisionBench converts ten source families (290 tasks; DNA, protein, RNA, and double sequences) into one typed-decision record format with a leak-proof admission protocol (Section 3). On a 37-task matched slice we train the two comparable arms — a shared typed-decision scorer (#1) and per-task heads on the same encoder (#2) — with family-balanced interleaving, stabilized optimization, per-family dev early stopping, and three seeds; the frozen Qwen3-0.6B full-candidate likelihood (#3) serves as an off-the-shelf reference (Section 4). Section 5 reports the seed-averaged comparison and three operational tests of #1’s claimed value, plus training-dynamics and mechanism ablations.

## Contributions.

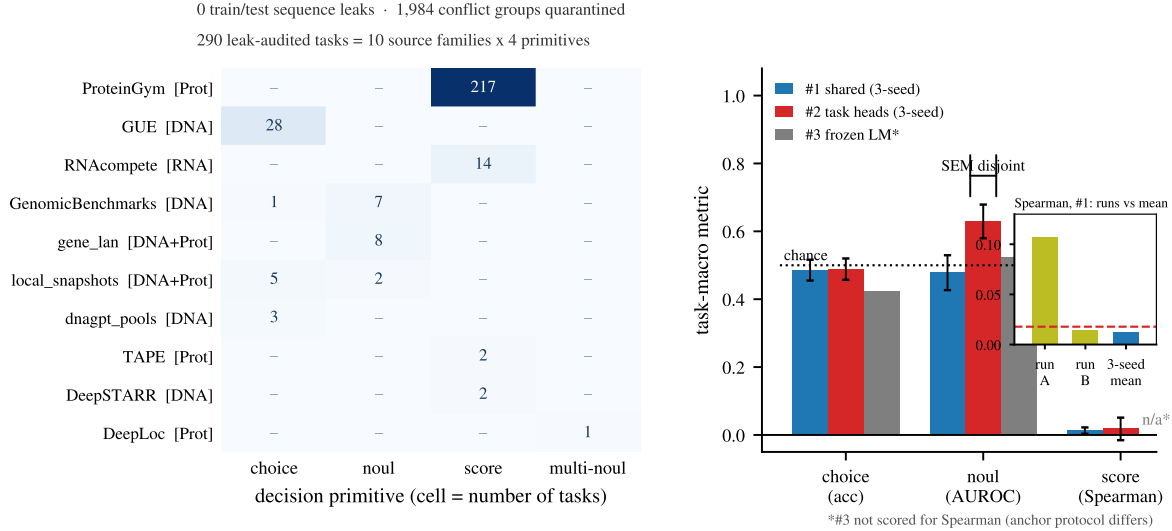
- **BioDecisionBench** (Section 3): a 290-task, three-modality, four-primitive biological decision benchmark with a leak-proof admission protocol — exact cross-split sequence isolation, conflict-group quarantine (1,984 groups, not majority-voted), verbatim preservation of legacy blind tests, and label mappings validated on full pool  $\cap$  test overlaps. Post-isolation within-task leakage is zero.
- **A seed-averaged evaluation protocol with a documented failure of single runs** (Sections 4.3 and 5.1): same-configuration reruns differ by up to  $7\times$  on score Spearman (0.107 vs 0.015) and reverse the comparison outcome; we make  $\geq 3$  seeds, per-task seed means, SEM + run-std reporting, and dev early stopping the minimum bar for this class of comparison.
- **The first matched-compute three-way comparison of decision architectures on biology** (Sections 5.1 and 5.2): per-task heads  $\geq$  shared scorer (Noul significantly better, Choice/Score at parity, and more stable across seeds); both trained arms beat the frozen candidate-likelihood baseline. The shared scorer’s advantage is *qualitative* — it can serve arbitrary new tasks with zero adaptation, but its zero-shot quality equals the untrained base, cross-family transfer is weak and classification-only, and  $\approx 32$  labels let a fresh head catch its zero-shot level.
- **Training-dynamics and mechanism diagnostics** (Sections 5.3 and 5.4): naive budget scaling destabilizes joint training below the untrained baseline; stabilization recovers but confirms a plateau; per-family checkpoint selection is a zero-inference-cost improvement (family-optimal steps span  $26\times$ ); two reusable controlled ablations (noul candidate phrasing across three stable checkpoints; contrastive decoding for frozen VLM candidate likelihood).

**Results preview.** Seed-averaged, per-task heads reach Noul AUROC 0.629 against the shared scorer’s 0.478 (SEMs disjoint) — while a single earlier run had suggested the opposite ordering. The shared scorer’s zero-shot interface, its headline feature, is caught by a fresh 32-label linear head on 7 of 11 held-out tasks.

Figure 1 summarizes the benchmark and the headline comparison.

## 2 Related Work

**Biological sequence benchmarks and foundation models.** DNA benchmarks cover histone marks, promoters, splice sites and regulatory annotations (GUE/DNABERT-2, 28 tasks (Zhou et al., 2024); Genomic Benchmarks (Gresova et al., 2022); DeepSEA (Zhou and Troyanskaya, 2015); DeepSTARR (de Almeida et al., 2022)); protein benchmarks cover fluorescence, stability, homology and mutation-fitness assays (TAPE (Rao et al., 2019); ProteinGym’s 217 DMS assays (Notin et al., 2023); FLIP (Dallago et al., 2021)); localization and function benchmarks are multi-label (DeepLoc 2.0 (Thummuluri et al., 2022); DeepGOZero (Kulmanov and Hoehndorf, 2022)); RNA benchmarks cover RBP binding preference (RNAcompete (Ray et al., 2013)). Foundation models — ESM-style protein LMs (Lin et al., 2023), protein language tooling (Xu et al., 2023b,a), and multimodal DNA agents (de Almeida et al., 2025) — are trained and ranked per family. These resources are family-siloed: incompatible



**Figure 1: BioDecisionBench and its headline comparison.** *Left:* 290 leak-audited tasks over 10 source families and 4 decision primitives; the admission protocol leaves zero within-task train/test sequence overlap (1,984 conflicting-label groups quarantined rather than majority-voted). *Right:* seed-averaged (3 seeds), matched-compute comparison per primitive (mean $\pm$ SEM over tasks). Per-task heads (#2) beat the shared typed-decision scorer (#1) on noul (disjoint SEMs) and tie on choice/score; both trained arms exceed the frozen candidate-likelihood baseline (#3, not compute-matched). *Inset:* the methodological warning — two single runs of the same configuration give score Spearman 0.107 vs 0.015 for #1; the 3-seed mean is 0.013.

record formats, split protocols and metric conventions make cross-family, cross-interface measurement impossible, and most are single-sequence, single-task-type. BioDecisionBench is not another leaderboard substrate: it converts these sources into one typed-decision record format with leakage auditing as a first-class citizen (Section 3.3) and serves as the evaluation base for the architecture comparison.

**Typed and candidate-scoring decision interfaces.** The Jev line of systems (Almeida, 2026) exposes structured Noul/Choice/Score probabilistic judgements behind a service interface (weights not public). Open variants include GLiClass, a ModernBERT uni-encoder with dynamic label sets (Stepanov et al., 2025); NanoJev, a small shared-scorer reimplement (TianyuCodings, 2026); and SemIf, which reads option logits from a frozen LLM (TheoLeeCJ, 2025). Constrained decoding is the generative counterpart (Scholak et al., 2021). Each of these validates the interface’s feasibility; none compares *shared scorer vs. per-task heads* under matched compute, same data, and multiple seeds. We report the honest, counter-intuitive outcome of exactly that comparison.

**Evaluation methodology and variance.** Repeated runs and variance reporting are standard advice in ML evaluation (Bouthillier et al., 2021; Miller, 2024); LLM-evaluation work documents sensitivity to phrasing, prompt format and option order (Miller, 2024). In biology, prior work measured candidate-order sensitivity in the same decision setting: reordering candidates changes  $\approx 9\%$  of predictions (Wang, 2026). This paper adds a concrete *conclusion-reversal* case for architecture comparison: two runs of one configuration give score Spearman 0.107 vs 0.015 for the shared scorer, flipping the winner; three-seed averaging overturns the single-run “shared  $\geq$  heads” verdict (Section 5.1). To our knowledge this is the first fully documented instance of run variance reversing a decision-architecture conclusion, and we operationalize the remedy:  $\geq 3$  seeds, per-task seed means, SEM and run-std reported separately, and dev early stopping with  $\geq 256$  dev samples.

**Probing and adaptation cost.** Linear probing and few-shot head fitting are standard tools for characterizing representations. We operationalize the “zero-adaptation interface” claim as a *catch-up cost*: freeze the encoder, fit a new head for a held-out task with  $N \in \{32, 128, 512\}$  labels, and report the smallest  $N$  at which the head reaches the shared scorer’s zero-shot level (Section 5.2). This turns “interface flexibility” from a qualitative claim into a comparable number.

**Table 1:** The four decision primitives, their extra fields, supervision and metrics.

| Primitive  | Extra fields                                                                                     | Supervision / evaluation                                                           |
|------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| choice     | candidates (mutually exclusive labels), answer                                                   | CE; accuracy, macro-F1, NLL, candidate-validity rate                               |
| noul       | candidates = {no, yes}, answer                                                                   | binary CE; AUROC, MCC, accuracy, calibration                                       |
| score      | score_levels (5), anchors (5 values from 4 interior train quantile cuts), gold_value, gold_level | soft-target CE + expected-value MSE; Spearman (primary), train-std-normalized RMSE |
| multi_noul | candidates (all labels), per-label 0/1 vector, answer = positive set                             | masked binary; micro/macro-F1, per-label calibration                               |

**Table 2:** BioDecisionBench composition. Rows are the 10 source families (canonical task-id prefix map used by the training scheduler; `lg_/protein_homology` share the `local_snapshots` family). Task counts are per evaluation unit; ProteinGym’s 217 assays share one question type. Admission: 0 train/test sequence leaks after isolation; 1,984 conflicting-label groups quarantined (not majority-voted); 925,985 cross-task duplicate sequences kept isolated per task.

| Family            | Mod.            | choice | noul | score | multi | Tasks | Train     | Dev     | Test    |
|-------------------|-----------------|--------|------|-------|-------|-------|-----------|---------|---------|
| ProteinGym        | P               | –      | –    | 217   | –     | 217   | 415,201   | 140,300 | 140,810 |
| GUE               | D               | 28     | –    | –     | –     | 28    | 731,069   | 82,350  | 82,125  |
| RNAcompete        | R               | –      | –    | 14    | –     | 14    | 275,449   | 33,610  | 33,476  |
| GenomicBenchmarks | D               | 1      | 7    | –     | –     | 8     | 516,071   | 57,251  | 162,541 |
| gene_lan          | D+P             | –      | 8    | –     | –     | 8     | 104,130   | 12,897  | 12,973  |
| local_snapshots   | D+P             | 5      | 2    | –     | –     | 7     | 69,469    | 6,894   | 7,048   |
| dnagpt_pools      | D               | 3      | –    | –     | –     | 3     | 111,269   | 12,527  | 13,892  |
| TAPE              | P               | –      | –    | 2     | –     | 2     | 74,955    | 7,874   | 40,042  |
| DeepSTARR         | D               | –      | –    | 2     | –     | 2     | 804,576   | 81,140  | 82,372  |
| DeepLoc           | P               | –      | –    | –     | 1     | 1     | 16,878    | 5,462   | 5,963   |
| <b>Total</b>      | D42+P226+R14+8D | 37     | 17   | 235   | 1     | 290   | 3,119,067 | 440,305 | 581,242 |

### 3 BioDecisionBench: Design and Construction

#### 3.1 Unified interface and record format

Every example is one JSON line carrying `task_id`, `primitive`, `modality`, `sequences` (each `{role, modality, sequence}`), `question`, `split`, `group`, and `provenance`, plus interface fields determined by the decision primitive  $p \in \{\text{noul}, \text{choice}, \text{score}, \text{multi\_noul}\}$  (Table 1). Sequences keep role boundaries for double-sequence tasks (`query_1/query_2`: DNA+protein, DNA+DNA, protein+protein) rather than being concatenated. `provenance` records source repository, revision/commit and license; download receipts include per-file SHA-256.

#### 3.2 Sources, scale, and modalities

Table 2 lists the composition. ProteinGym contributes 217 mutation-fitness assays using the official contiguous-5 per-mutant folds (fold 0 = test, fold 1 = dev, folds 2–4 = train; multi-mutant rows excluded); GUE 28 DNA choice tasks; RNAcompete 14 RNA score tasks (14-RBP subset, probes subsampled to 25k with a fixed seed); Genomic Benchmarks 8; `gene_lan_transfer` 8 double-sequence noul tasks with endpoint union-find component grouping; `dnagpt` DNA pools 3 (+1 promoter); TAPE 2; DeepSTARR 2 (dual outputs split into two tasks); DeepLoc 2.0 as one multi-noul task; and 7 local snapshots from earlier public releases (Wang, 2024, 2026). Interface distribution: choice 37 / noul 17 / score 235 / multi\_noul 1; modality: DNA 42 / protein 226 / RNA 14 / double-sequence 8; rows: 3,119,067 train / 440,305 dev / 581,242 test.

*Counting convention.* The 290 figure counts evaluation units, not question types: ProteinGym’s 217 assays share one question type. We report both numbers wherever the benchmark scale is quoted and never claim “290 independent questions”.

#### 3.3 Leak-proof admission protocol

1. **Native splits first.** Sources with official train/dev(/test) splits are used verbatim (GUE, Genomic Benchmarks, TAPE, DeepSTARR, ProteinGym folds, local snapshots).

2. **Legacy blind tests isolated from upstream pools.** The dnagpt pools (splice, TF, core) ship as unsplit “train” data *containing the blind-test members of prior studies* (Wang, 2026). Handling: test = legacy blind test, kept verbatim; train/dev = the pool with *all* legacy-test members removed, bucketed by sequence hash; rows whose pool integer label conflicts with the blind-test text label (8 in core) are quarantined as a group. Label mappings are not guessed but validated on full pool $\cap$ test overlaps (TF 3437/3437; core 5910/5918; splice agrees 4545/4545 under the legacy convention 0 $\rightarrow$ Non-Splice/1 $\rightarrow$ Acceptor/2 $\rightarrow$ Donor, while the literature’s newer builder uses a different convention — we adopt the data-consistent one and flag the acceptor/donor semantics for independent verification).
3. **Exact cross-split sequence isolation.** A sequence appearing in several splits is assigned wholly to the highest-priority split (test > dev > train); lower-split copies are dropped. Same-sequence conflicting labels quarantine the whole group (1,984 groups; virus\_covid contributes 1,754) rather than majority-voting. We do *not* merge reverse complements: RC equivalence flips labels for strand-sensitive tasks (splice donor/acceptor, promoter direction); label-aware RC grouping is left as future work.
4. **Double-sequence tasks grouped by endpoints.** gene\_lan pairs are split by union-find connected components over endpoints, so no endpoint appears in both train and test. A construction-diagnosable subset (the two “random” pairing tasks are separable by a frame-0 codon-table translation) is declared in the task notes and the paper, and is not claimed as an independent blind test.
5. **Score anchor discipline.** The 5-level anchors are derived from training data only (4 interior quantile cuts  $\rightarrow$  5 representative values); test values never feed back. Raw RMSE is never averaged across assays (native scales differ by orders of magnitude); Spearman is the primary metric.
6. **Admission checker.** Per task we verify split overlap, label conflicts, answer  $\in$  candidates, and length distributions; globally we count cross-task duplicate sequences (925,985 sequence keys occur in >1 task — not within-task leakage under a multi-task model, but they must be escalated to global isolation under leave-one-task-out). Final admission result: **0 within-task leaks, 0 flagged tasks.**

### 3.4 Scope boundaries

Under the current encoder path (single sequence, DNA/protein, 512-token budget), 241/290 tasks are consumable by the compared architectures; the remaining 49 (14 RNA, 10 double-sequence, 1 multi-noul, 24 over-budget) are in the benchmark, covered by the corresponding modality/interface evaluation paths or reported with exclusion rates — never silently truncated. The benchmark’s value is the protocol and the data; it is not bound to any single evaluated architecture.

## 4 Evaluation Framework

### 4.1 The three compared architectures

#1 implements the typed-decision ideal: candidates are dynamic text, the scorer weights are shared across all tasks, and a new task needs no parameters. #2 is the conventional alternative on the same encoder. #3 answers “what does an off-the-shelf small LM get, for free?” and is reported for reference only.

### 4.2 Training protocol

Joint training interleaves families: each family contributes an equal number of batches per epoch (rotating within family), so ProteinGym’s 217 assays cannot flood the diverse small families. AdamW with encoder lr  $2 \times 10^{-5}$ , head/scorer lr  $1 \times 10^{-4}$ ; warmup fraction 0.15 and gradient clipping 0.5 — values set by the instability diagnosis of Section 5.3, not cherry-picked post hoc; cosine decay to 10%; batch 64 with gradient accumulation; 512-token full inputs (over-budget examples are excluded and the exclusion rate reported). Checkpoint selection: **per-family dev early stopping** ( $\geq 256$  dev samples per family; each family keeps its own best-dev checkpoint from the same trajectory — zero inference-time cost), motivated by the  $26\times$  spread of family-optimal steps (Section 5.3).

**Table 3:** Compared decision architectures. #1 and #2 are matched-compute (same data, updates, optimizer, seeds; the only variable is the decision mechanism). #3 is *not* matched (frozen, different backbone, 120-character sequence truncation, different score protocol) and is labeled as such wherever it appears.

|                 | #1 shared typed-decision scorer                                                                                                                                        | #2 encoder + per-task heads                                                                  | #3 frozen candidate likelihood                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------|
| backbone        | Laya ModernBERT-large (Warner et al., 2025; NandhaKishorM, 2026), 423M, biology-expanded BPE (52,412)                                                                  | same as #1                                                                                   | Qwen3-0.6B (Yang et al., 2025), 596M causal LM   |
| decision        | 2-layer typed transformer + shared scalar scorer over candidate markers; choice pools over a candidate-set attention; score emits a 5-level distribution + expectation | one Linear head per task: choice/noul = fixed-class softmax; score = scalar regression (MSE) | length-normalized logprob per candidate, arg max |
| task parameters | none (candidates are input text)                                                                                                                                       | one $ \mathcal{Y}_t $ -way head per task                                                     | none (frozen)                                    |
| new task        | inference with zero adaptation                                                                                                                                         | requires a new head + labeled training                                                       | inference with zero adaptation                   |
| training        | matched budget                                                                                                                                                         | matched budget                                                                               | none                                             |

**Table 4:** Matched-compute three-way comparison, seed-averaged over 3 seeds. Mean $\pm$ SEM across tasks after per-task seed averaging; run-std is the across-seed standard deviation of the task-macro aggregate. #3 (frozen Qwen3-0.6B candidate likelihood) is *not* compute-matched and is reported for reference.

| Primitive                               | Metric   | $n$ | #1 shared         | #2 task heads     | #3 frozen LM | run-std #1 | run-std #2  |
|-----------------------------------------|----------|-----|-------------------|-------------------|--------------|------------|-------------|
| choice                                  | accuracy | 21  | 0.486 $\pm$ 0.031 | 0.489 $\pm$ 0.031 | 0.424        | 0.040      | 0.030       |
| noul                                    | AUROC    | 7   | 0.478 $\pm$ 0.051 | 0.629 $\pm$ 0.050 | 0.522        | 0.087      | 0.030       |
| score                                   | Spearman | 9   | 0.013 $\pm$ 0.009 | 0.018 $\pm$ 0.033 | —            | 0.064      | 0.054       |
| Per-task seed-mean wins (#1 / tie / #2) |          |     |                   |                   |              |            | 4 / 23 / 10 |

### 4.3 Seed-averaging protocol

Motivating case (Section 5.1): two runs of the same configuration and seed give score Spearman 0.107 vs 0.015 for #1, reversing the comparison. Protocol:  $\geq 3$  seeds (20261001/02/03); average per task across seeds *first*, then aggregate across tasks; report mean $\pm$ SEM (across tasks) and run-std (across seeds) separately; early-stop dev sets  $\geq 256$  to shrink selection noise; any single-run number is labeled as such. GPU nondeterminism (identical-seed trajectories diverge after  $\sim 137$  steps) is the physical source and cannot be removed by fixing seeds — only absorbed by averaging.

### 4.4 Metric conventions

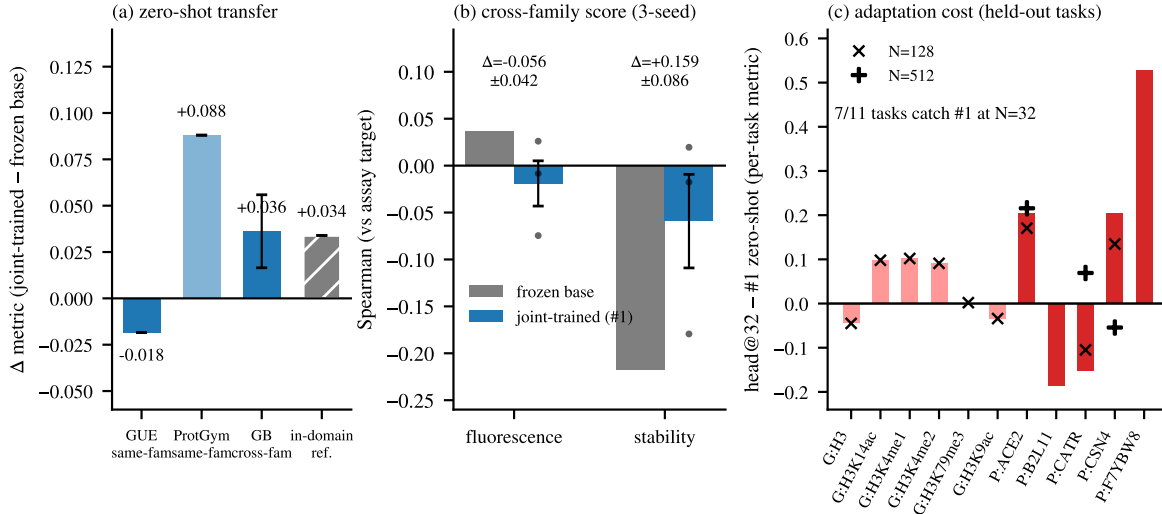
choice: accuracy (primary) + macro-F1. noul: AUROC (primary — ranking-based, immune to threshold and calibration) + MCC + accuracy. score: Spearman (primary, scale-free) + RMSE normalized by train std; **raw RMSE is never averaged across assays**. multi\_noul: per-label sigmoid protocol (prepared in the benchmark; not consumed by the encoder path in this paper) (calibration protocol follows Guo et al., 2017).

## 5 Results

### 5.1 Main comparison: seed-averaged three-way

The matched slice has 37 tasks (21 choice, 7 noul, 9 score; ProteinGym and GUE capped at 12 each), 2,520 updates, 3 seeds, per-family early stopping. Table 4 and the right panel of Figure 1 give the result.

**Reading.** (i) #2  $\geq$  #1: noul is significantly better (AUROC 0.629  $\pm$  0.050 vs 0.478  $\pm$  0.051, disjoint SEMs), choice and score are at parity (0.489 vs 0.486; 0.018 vs 0.013), and #2 is more stable across seeds (noul run-std



**Figure 2: Three operational tests of the shared scorer’s claimed advantages.** (a) Zero-shot transfer to same-family held-out tasks is a wash (GUE  $-0.018$ , ProtGym  $+0.088$ ); cross-family transfer on GenomicBenchmarks is weakly positive ( $+0.036$ , SEM over 3 seeds), comparable to the in-domain reference ( $+0.034$ ; full2 configuration, not strictly matched). (b) Cross-family *score* transfer (TAPE, 3 seeds) does not hold: fluorescence degrades ( $\Delta = -0.056 \pm 0.042$ ); stability’s apparent gain ( $+0.159 \pm 0.086$ ) is regression of a pathological frozen baseline ( $-0.218$ ) toward zero — trained absolute Spearman stays negative ( $-0.059$ ). (c) Adaptation cost: a fresh #2 linear head catches #1’s zero-shot on 7/11 held-out tasks with only  $N=32$  labels ( $\times$ :  $N=128$ ,  $+$ :  $N=512$ ).

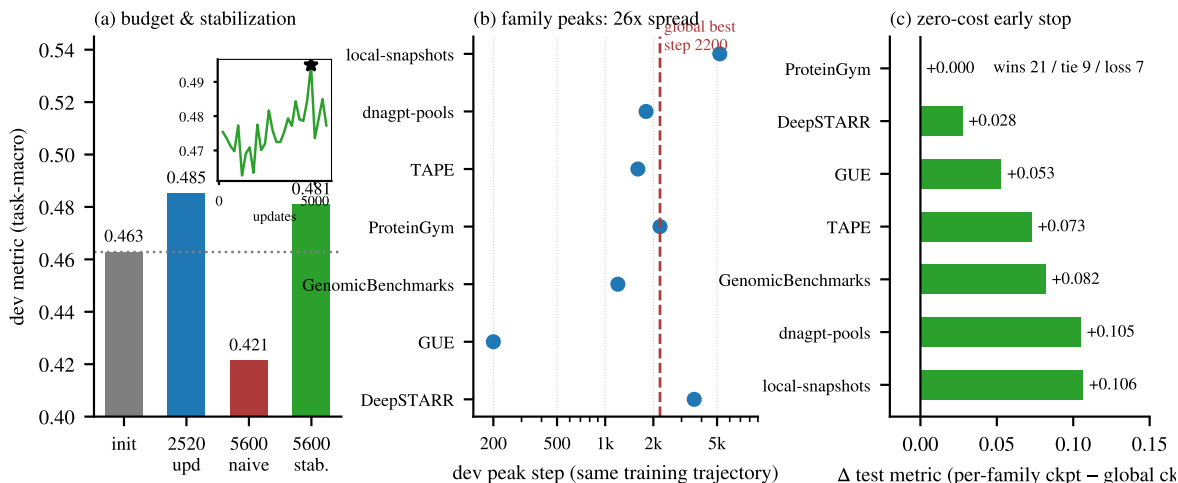
0.030 vs 0.087; also 0.030 vs 0.040 on choice, 0.054 vs 0.064 on score). Per-task seed-mean wins: #1 4 / ties 23 / #2 10. (ii) Both trained arms beat the frozen #3 on choice (0.49/0.49 vs 0.42) and on noul for #2 (0.629 vs 0.522) — but #1 *loses* to the frozen small LM on noul (0.478 vs 0.522): a trained shared scorer ranks binary candidates worse than an untrained model reading their likelihoods. (iii) Absolute levels are modest (choice  $\approx 0.49$ , near majority-class baselines): this is the joint-37, 2,520-update, 512-window stress regime, calibrated against specialized-training anchors in Section 6.

**The variance case.** A single earlier run (before the seed protocol existed) reported #1 score Spearman 0.107 vs #2  $-0.007$  and a “#1  $\geq$  #2” overall verdict. A rerun of the identical configuration and seed fell to 0.015 — on *score* tasks, which do not touch the only code difference between the runs (noul candidate phrasing), so the gap is pure run variance; per-family early-stop steps drifted (GUE 200 $\rightarrow$ 5000, GB 1200 $\rightarrow$ 200). With 3-seed averaging, both arms converge to  $\approx 0.01$ – $0.02$  and the “#1 clearly wins score” claim evaporates. This is the concrete failure mode Section 4.3 protocolizes against.

## 5.2 Three operational tests of #1’s claimed value

**(a) Same-family zero-shot.** Held-out tasks from families already in training (16 GUE + 20 ProteinGym tasks unseen by the 37-task joint model): #1-trained  $\approx$  #1-frozen base (GUE accuracy 0.496 vs 0.515,  $\Delta = -0.018$ ; per-task 11 wins / 9 ties / 13 losses — a wash). Joint training on 37 tasks did not learn a strongly transferable decision function. #2 cannot participate at all: it has no head for unseen label sets — the qualitative difference stands, the quantitative one does not.

**(b) Cross-family zero-shot.** Holding out the entire GenomicBenchmarks family (training on the other six families) and evaluating its 8 tasks gives mean  $\Delta = +0.036 \pm 0.034$  over 3 seeds (per-seed  $+0.038/-0.006/+0.077$ ) — the same magnitude as directly training on GB in-domain ( $+0.034$ ), hinting the gain is a generic decision improvement rather than family-specific fitting, but the std matches the mean and per-task effects range  $-0.19$  to  $+0.31$ . For the *score* interface, holding out TAPE (while training on the DeepSTARR and ProteinGym score families) over 3 seeds shows no clean transfer: fluorescence degrades ( $\Delta = -0.056 \pm 0.042$ , negative in all three seeds), and stability’s apparent  $+0.159 \pm 0.086$  is entirely regression of a pathological frozen baseline ( $-0.218$ ) toward zero — the trained absolute Spearman remains negative ( $-0.059$ ). Cross-family transfer exists, weakly, for



**Figure 3: Training dynamics under joint multi-task fitting.** (a) Naive budget scaling destabilizes: 5,600 updates score *below* init (0.421) while 2,520 reaches 0.485; stabilization (warmup 0.15, grad clip 0.5, dev early stop) recovers 0.481 at 5,600 — a plateau, not overfitting (inset: stabilized dev curve; star = selected checkpoint). (b) Family-optimal dev peaks on the *same* trajectory span steps 200–5,200 (26×), while the global best sits at 2,200. (c) Per-family early stopping (zero inference-time cost) dominates at family level (all  $\Delta \geq 0$ , mean +0.063); per task: 21 win / 9 tie / 7 loss vs the global checkpoint.

classification only.

**(c) Adaptation cost.** Freezing #2’s trained encoder and fitting a fresh linear head for each of 11 held-out tasks (6 GUE + 5 ProteinGym) with  $N$  labels: the new head catches or beats #1’s zero-shot on **7/11 tasks at  $N = 32$**  (Table 6); #1 stays ahead on 3/11 even when #2 is given  $N = 512$ ; one task needs  $N = 512$ . Family means: GUE 0.476 (zero-shot) vs 0.511 (head@32); ProteinGym  $-0.024$  vs 0.095. Thirty-two labels is a trivial cost — the “zero-adaptation” selling point does not convert into a meaningful performance-or-cost advantage.

**(d) Noul phrasing on stable checkpoints.** Candidate text is an inference-time degree of freedom that #1 has and #2 cannot have. A controlled ablation (same checkpoint, only the rendered candidate text varies; six phrasings) on the earlier single checkpoint had suggested raw “no/yes” beats the redundant “false: no / true: yes” by +0.066 AUROC. Re-running on three retrained seed-averaged stable checkpoints (Figure 4a) *reverses* the direction: the redundant prefix form wins on all three seeds ( $0.553 \pm 0.023$  vs the trained phrasing’s  $0.508 \pm 0.024$ ), and **no phrasing reaches #2** (best 0.553 vs 0.629). Two conclusions: the noul deficit of #1 is not a phrasing artifact (the main result is reinforced), and even inference-level “clean controlled” evidence is checkpoint-dependent — ablations need multi-checkpoint confirmation too.

### 5.3 Training dynamics

**Budget scaling is negative.** On the same 43-task set, doubling updates 2,520→5,600 (and per-task data 768→2048) drops classification accuracy 0.485→0.421 — *below* the untrained init (0.463) — and TAPE Spearman  $0.237 \rightarrow -0.020$ . Diagnosis: training loss plateaus at  $\sim 1.06$  after  $\sim 900$  steps, gradient median stays  $\approx 0.25$  for 4,700 further steps with early spikes (max 547), and fold classification never leaves chance (train loss  $\approx \ln 7$  throughout). This is joint-optimization instability landing in bad basins, not overfitting: the training loss never approaches zero.

**Stabilization repairs, and confirms a plateau.** Warmup 0.05→0.15, clip 1.0→0.5, and dev early stopping (every 200 steps, best-dev promotion) recover the 5,600-update budget to 0.481 — statistically the small budget’s 0.485 (gain vs init +0.049  $\approx$  +0.050). Beyond  $\approx 2,500$  updates, compute is not the bottleneck; the ceiling is capacity/interference of joint multi-task training (Figure 3a).

**Per-family early stopping is free.** On one training trajectory, family-optimal dev peaks span steps 200 (GUE) to 5,200 (local snapshots) — a 26× spread — while the global best sits at 2,200. Selecting each family’s own

best-dev checkpoint dominates the single global checkpoint at family level (all  $\Delta \geq 0$ ; mean +0.063; per task 21/9/7 wins/ties/losses; Figure 3b,c), at zero inference-time cost. Part of the plateau was a checkpoint-selection compromise, not pure capacity.

## 5.4 Controlled mechanism ablations

Beyond the phrasing ablation of Section 5.2d (candidate surface form is a real but run-dependent degree of freedom, and no phrasing closes the noul gap), one more probe concerns frozen candidate-likelihood scoring: on a multimodal route-A pilot (synthetic property maps, frozen Qwen2.5-VL (Bai et al., 2025)), plain candidate likelihood collapses to one class (accuracy  $0.040 < \text{chance}$ ), while *contrastive decoding*  $\log p(\text{cand} \mid \text{img}, q) - \log p(\text{cand} \mid q)$  (cf. Leng et al., 2024; Li et al., 2023) restores multi-class predictions but stays at chance (Figure 4b). The reusable lesson: frozen-VLM candidate-likelihood evaluation should default to contrastive decoding; the route itself is rejected (Appendix C).

## 6 Discussion and Limitations

**When would a shared typed-decision scorer win?** Within our evidence: not on seen tasks, not zero-shot (at base level), weakly on cross-family classification only, and its zero-adaptation edge is erased by  $\approx 32$  labels. What survives unambiguously is the interface itself: no task parameters, arbitrary dynamic candidates, immediate inference on new tasks (#2 architecturally cannot), and inference-time degrees of freedom (phrasing, per-family checkpoints) — though the 3-seed ablation shows phrasing tuning does not close the noul gap ( $0.553 < 0.629$ ). Deployment regimes with very large or open candidate sets, frequent task arrivals, or true zero-label cold starts may shift the balance; those are hypotheses, not our results.

**Absolute levels vs. specialized anchors.** Joint-37 choice  $\approx 0.49$  and score  $\approx 0.01$ – $0.02$  sit far below specialized training: the same architecture family reaches 87–89% on promoters, 56–58% on fold class and GFP MAE 0.32–0.49 under per-task recipes (Wang, 2024). The joint regime is a deliberate stress test (one checkpoint serving 37 heterogeneous tasks,  $\leq 2048$  samples each,  $\approx 70$  updates per task); it calibrates “where shared decision interfaces currently stand under multi-task small budgets” and is not an architecture ceiling.

**Threats to validity.** (i) The head-to-head uses a 37-task slice (PG/GUE capped at 12), not all 290: the benchmark is the infrastructure contribution and the comparison is a matched slice — we state the two scopes separately everywhere. (ii) 3 seeds; SEM is across tasks, run-std across seeds; noul ( $n=7$ ) and score ( $n=9$ ) have few tasks. (iii) #3 is not matched (frozen, different backbone, truncation, different score protocol). (iv) The cross-family score arm now has 3 seeds, but with  $n=2$  tasks the stability  $\Delta = +0.159$  depends on the pathological frozen baseline and must be read with that caveat (Section 5.2). (v) dev selection noise is mitigated (96→256 samples) but not eliminated. (vi) GPU nondeterminism is irreducible; only seed averaging absorbs it. (vii) the 512-token budget excludes long-sequence examples (exclusion rates reported, never silent truncation).

## 7 Conclusion

We introduced BioDecisionBench — 290 biological sequence decision tasks from 10 source families, three modalities and four typed decision primitives, under one record format and a leak-proof admission protocol (zero within-task leakage after isolation; conflict groups quarantined, legacy blind tests preserved verbatim) — together with reproducible build/evaluation/training code. On its 37-task matched slice, the first seed-averaged three-way architecture comparison gives a bounded, honest answer: the shared typed-decision scorer does not beat per-task heads (significantly worse noul, parity elsewhere, less stable), its zero-shot and cross-family transfer are weak, and its zero-adaptation claim is matched by a 32-label fresh head. Its real value is the interface: dynamic candidates, no task parameters, immediate new-task inference. Methodologically, we documented a complete conclusion-reversal case from run variance ( $\pm 0.05$ – $0.1$ ) and propose seed averaging, run-std reporting and stabilized selection as the minimum bar for architecture comparisons in this regime. Future work: per-family training over all 290 tasks, per-task early stopping, a matched generative arm (fine-tuned small LM), real-image multimodal evaluation, and re-testing transfer at larger budgets and backbones.

## References

- Diogo Almeida. Introducing system one models & Jev. TypeSafe AI technical announcement, September 2026. URL <https://typesafe.ai/blog/introducing-system-one-models-and-jev>. Published 15 September 2026; accessed 24 September 2026. Product documentation, not a peer-reviewed biological evaluation.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025.
- Xavier Bouthillier, Pierre Delaunay, Mirko Bronzi, Assya Trofimov, Brennan Nichyporuk, Justin Szeto, Naz Sepah, Edward Raff, Kanika Madan, Vikram Voleti, Samira Ebrahimi Kahou, Vincent Michalski, Dmitriy Serdyuk, Tal Arbel, Chris Pal, Gaël Varoquaux, and Pascal Vincent. Accounting for variance in machine learning benchmarks, 2021.
- Christian Dallago, Jody Mou, Kadina E. Johnston, Bruce J. Wittmann, Nicholas Bhattacharya, Samuel Goldman, Ali Madani, and Kevin K. Yang. Flip: Benchmark tasks in fitness landscape inference for proteins. *bioRxiv*, November 2021. doi: 10.1101/2021.11.09.467890. URL <http://dx.doi.org/10.1101/2021.11.09.467890>.
- Bernardo P. de Almeida, Franziska Reiter, Michaela Pagani, and Alexander Stark. Deepstarr predicts enhancer activity from dna sequence and enables the de novo design of synthetic enhancers. *Nature Genetics*, 54(5): 613–624, May 2022. ISSN 1546-1718. doi: 10.1038/s41588-022-01048-5. URL <http://dx.doi.org/10.1038/s41588-022-01048-5>.
- Bernardo P. de Almeida, Guillaume Richard, Hugo Dalla-Torre, Christopher Blum, Lorenz Hexemer, Priyanka Pandey, Stefan Laurent, Chandana Rajesh, Marie Lopez, Alexandre Laterre, Maren Lang, Uğur Şahin, Karim Beguir, and Thomas Pierrot. A multimodal conversational agent for dna, rna and protein tasks. *Nature Machine Intelligence*, 7(6):928–941, June 2025. ISSN 2522-5839. doi: 10.1038/s42256-025-01047-1. URL <http://dx.doi.org/10.1038/s42256-025-01047-1>.
- Katarina Gresova, Vlastimil Martinek, David Cechak, Petr Simecek, and Panagiotis Alexiou. Genomic benchmarks: A collection of datasets for genomic sequence classification. *bioRxiv*, June 2022. doi: 10.1101/2022.06.08.495248. URL <http://dx.doi.org/10.1101/2022.06.08.495248>.
- Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In Doina Precup and Yee Whye Teh, editors, *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*, pages 1321–1330. PMLR, 06–11 Aug 2017. URL <https://proceedings.mlr.press/v70/guo17a.html>.
- Maxat Kulmanov and Robert Hoehndorf. DeepGOZero: improving protein function prediction from sequence and zero-shot learning based on ontology axioms. *Bioinformatics*, 38(Suppl. 1):i238–i245, 2022. doi: 10.1093/bioinformatics/btac256. URL <https://doi.org/10.1093/bioinformatics/btac256>.
- Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, page 13872–13882. IEEE, June 2024. doi: 10.1109/cvpr52733.2024.01316. URL <http://dx.doi.org/10.1109/cvpr52733.2024.01316>.
- Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 12286–12312. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.acl-long.687. URL <http://dx.doi.org/10.18653/v1/2023.acl-long.687>.
- Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, March 2023. ISSN 1095-9203. doi: 10.1126/science.ade2574. URL <http://dx.doi.org/10.1126/science.ade2574>.

- Evan Miller. Adding error bars to evals: A statistical approach to language model evaluations, 2024.
- NandhaKishorM. Laya: Typed-decision encoder software. Software repository and model card, 2026. URL <https://github.com/NandhaKishorM/laya>. Source revision 573e5b62696ba441230cd6be71d593331b5d23af; accessed 2026-09-23.
- Pascal Notin, Aaron W. Kollasch, Daniel Ritter, Lood van Niekerk, Steffanie Paul, Hansen Spinner, Nathan Rollins, Ada Shaw, Ruben Weitzman, Jonathan Frazer, Mafalda Dias, Dinko Franceschi, Rose Orenbuch, Yarin Gal, and Debora S. Marks. Proteingym: Large-scale benchmarks for protein design and fitness prediction. *bioRxiv*, December 2023. doi: 10.1101/2023.12.07.570727. URL <http://dx.doi.org/10.1101/2023.12.07.570727>.
- Roshan Rao, Nicholas Bhattacharya, Neil Thomas, Yan Duan, Xi Chen, John Canny, Pieter Abbeel, and Yun S. Song. Evaluating protein transfer learning with TAPE. In *Advances in Neural Information Processing Systems*, 2019. URL <https://arxiv.org/abs/1906.08230>.
- Debashish Ray, Hilal Kazan, Kate B. Cook, Matthew T. Weirauch, Hamed S. Najafabadi, Xiao Li, Serge Gueroussov, Mihai Albu, Hong Zheng, Ally Yang, Hong Na, Manuel Irimia, Leah H. Matzat, Ryan K. Dale, Sarah A. Smith, Christopher A. Yarosh, Seth M. Kelly, Behnam Nabet, Desirea Mecnas, Weimin Li, Rakesh S. Laishram, Mei Qiao, Howard D. Lipshitz, Fabio Piano, Anita H. Corbett, Russ P. Carstens, Brendan J. Frey, Richard A. Anderson, Kristen W. Lynch, Luiz O. F. Penalva, Elissa P. Lei, Andrew G. Fraser, Benjamin J. Blencowe, Quaid D. Morris, and Timothy R. Hughes. A compendium of rna-binding motifs for decoding gene regulation. *Nature*, 499(7457):172–177, July 2013. ISSN 1476-4687. doi: 10.1038/nature12311. URL <http://dx.doi.org/10.1038/nature12311>.
- Torsten Scholak, Nathan Schucher, and Dzmitry Bahdanau. PICARD: Parsing incrementally for constrained auto-regressive decoding from language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 9895–9901. Association for Computational Linguistics, November 2021. doi: 10.18653/v1/2021.emnlp-main.779. URL <https://aclanthology.org/2021.emnlp-main.779/>.
- Ihor Stepanov, Mykhailo Shtopko, Dmytro Vodianytskyi, Oleksandr Lukashov, Alexander Yavorskyi, and Mykyta Yaroshenko. Gliclass: Generalist lightweight model for sequence classification tasks, 2025.
- TheoLeeCJ. Semif-openjev: Semantic ifs from open models. <https://github.com/TheoLeeCJ/SemIf-OpenJev>, 2025. Community reimplementaion, not affiliated with TypeSafe; accessed 2026-10-06.
- Vineet Thummuluri, José Juan Almagro Armenteros, Alexander Rosenberg Johansen, Henrik Nielsen, and Ole Winther. DeepLoc 2.0: multi-label subcellular localization prediction using protein language models. *Nucleic Acids Research*, 50(W1):W228–W234, 2022. doi: 10.1093/nar/gkac278. URL <https://doi.org/10.1093/nar/gkac278>.
- TianyuCodings. Nanojev: A nano replica of jev – parallel decisions, dynamic candidates, and an end-to-end typed-decision model. <https://github.com/TianyuCodings/NanoJev>, 2026. Community reimplementaion; accessed 2026-10-06.
- Liang Wang. LLaMA-Gene: A general-purpose gene task large language model based on instruction fine-tuning. *arXiv preprint arXiv:2412.00471*, 2024. doi: 10.48550/arXiv.2412.00471. URL <https://arxiv.org/abs/2412.00471>.
- Liang Wang. BioPAWS-2: A unified chat-form instruction-tuning dataset and benchmark for biological foundation models. Hugging Face dataset card, 2026. URL <https://huggingface.co/datasets/dnagpt/biopaws-2>. Dataset documentation; local experimental files identified by SHA-256.
- Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Griffin Thomas Adams, Jeremy Howard, and Iacopo Poli. Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference. In Wanxiang Che, Joyce Nabende, Ekaterina Shutova, and Mohammad Taher Pilehvar, editors, *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2526–2547, Vienna, Austria, July 2025. Association for Computational Linguistics. ISBN 979-8-89176-251-0. doi: 10.18653/v1/2025.acl-long.127. URL <https://aclanthology.org/2025.acl-long.127/>.

**Table 5:** Per-family, per-seed means for the two trained arms (task-macro of per-task seed means). #1 uses per-family early-stop checkpoints (37-task slice; families as in the training scheduler), #2 the full-run checkpoint. DeepLoc (multi\_noul) and double-sequence tasks are outside this harness.

| Family            | Primitive | #1 (mean $\pm$ std over 3 seeds) | #2                 | n |
|-------------------|-----------|----------------------------------|--------------------|---|
| DeepSTARR         | score     | 0.035 $\pm$ 0.027                | 0.053 $\pm$ 0.023  | 3 |
| GUE               | choice    | 0.464 $\pm$ 0.017                | 0.478 $\pm$ 0.013  | 3 |
| GenomicBenchmarks | choice    | 0.402 $\pm$ 0.082                | 0.305 $\pm$ 0.014  | 3 |
| GenomicBenchmarks | noul      | 0.478 $\pm$ 0.017                | 0.629 $\pm$ 0.005  | 3 |
| ProteinGym        | score     | 0.015 $\pm$ 0.058                | 0.022 $\pm$ 0.015  | 3 |
| TAPE              | score     | -0.014 $\pm$ 0.047               | -0.027 $\pm$ 0.043 | 3 |
| dnagpt_pools      | choice    | 0.519 $\pm$ 0.006                | 0.514 $\pm$ 0.030  | 3 |
| local_snapshots   | choice    | 0.534 $\pm$ 0.016                | 0.536 $\pm$ 0.030  | 3 |

Hanwen Xu, Addie Woicik, Hoifung Poon, Russ B. Altman, and Sheng Wang. Multilingual translation for zero-shot biomedical classification using biotranslator. *Nature Communications*, 14(1), February 2023a. ISSN 2041-1723. doi: 10.1038/s41467-023-36476-2. URL <http://dx.doi.org/10.1038/s41467-023-36476-2>.

Minghao Xu, Xinyu Yuan, Santiago Miret, and Jian Tang. ProtST: Multi-modality learning of protein sequences and biomedical texts. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett, editors, *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 38749–38767. PMLR, 23–29 Jul 2023b. URL <https://proceedings.mlr.press/v202/xu23t.html>.

An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiayi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report, 2025.

Jian Zhou and Olga G Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12(10):931–934, August 2015. ISSN 1548-7105. doi: 10.1038/nmeth.3547. URL <http://dx.doi.org/10.1038/nmeth.3547>.

Zhihan Zhou, Yanrong Ji, Weijian Li, Pratik Dutta, Ramana Davuluri, and Han Liu. Dnabert-2: Efficient foundation model and benchmark for multi-species genomes. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 41642–41665, 2024. URL [https://proceedings.iclr.cc/paper\\_files/paper/2024/file/b633e7052970b8f5aa1a69164d99e9e8-Paper-Conference.pdf](https://proceedings.iclr.cc/paper_files/paper/2024/file/b633e7052970b8f5aa1a69164d99e9e8-Paper-Conference.pdf).

## A Full per-family results

Table 5 reports per-family, per-seed means for the two trained arms.

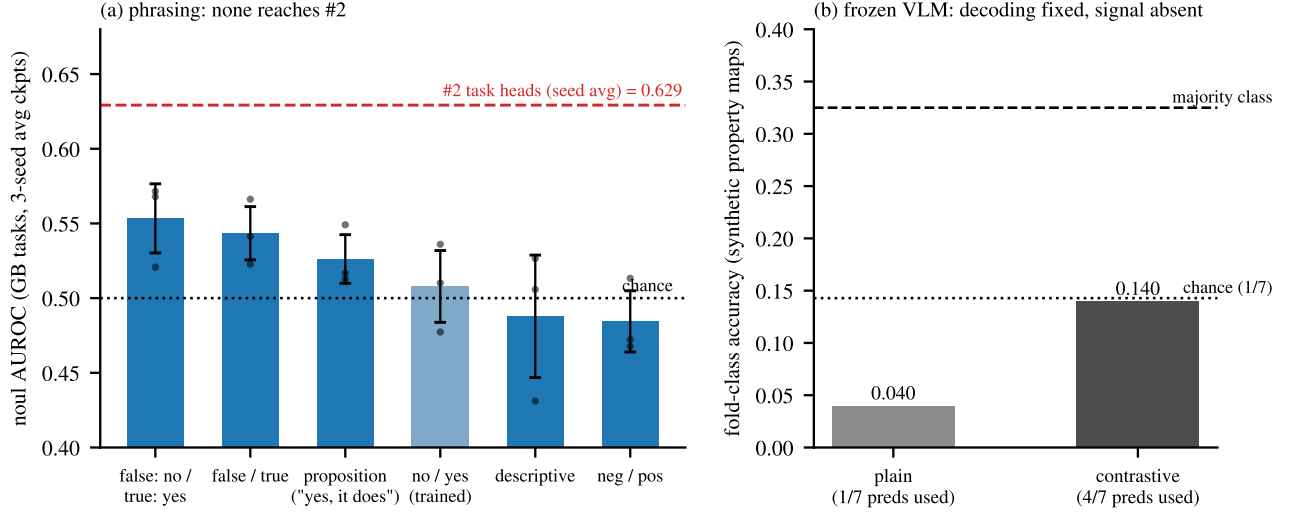
## B Adaptation cost and ablation tables

## C Multimodal route-A pilot

Protein sequences without resolved structures were rendered as synthetic amino-acid property heatmaps (7 normalized physicochemical tracks  $\times$  position, matplotlib viridis) and presented to a frozen Qwen2.5-VL-3B (Bai et al., 2025) alongside the fold-class question. Plain full-candidate likelihood collapsed to a single candidate on all 200 images (accuracy 0.040; macro-F1 0.011; 1/7 classes predicted) — candidate text priors dominate and the image does not change the ranking. Contrastive decoding,  $\log p(\text{cand} \mid \text{img}, q) - \log p(\text{cand} \mid q)$ , restores multi-class

**Table 6:** Adaptation cost on 11 held-out tasks: #1 zero-shot (no labels, no training) vs #2 new linear head fitted on  $N$  labels (frozen shared encoder). #2 catches or beats #1’s zero-shot on 7/11 tasks already at  $N=32$ .

| Task                                    | Metric   | #1 zero-shot | #2 $N=32$ | $N=128$ | $N=512$ |
|-----------------------------------------|----------|--------------|-----------|---------|---------|
| gue_H3                                  | accuracy | 0.555        | 0.510     | 0.510   | –       |
| gue_H3K14ac                             | accuracy | 0.445        | 0.543     | 0.543   | –       |
| gue_H3K4me1                             | accuracy | 0.430        | 0.532     | 0.532   | –       |
| gue_H3K4me2                             | accuracy | 0.450        | 0.541     | 0.541   | –       |
| gue_H3K79me3                            | accuracy | 0.490        | 0.492     | 0.492   | –       |
| gue_H3K9ac                              | accuracy | 0.485        | 0.451     | 0.451   | –       |
| pg_ACE2_HUMAN_Chan_2020                 | spearman | -0.176       | 0.029     | -0.005  | 0.040   |
| pg_B2L11_HUMAN_Dutta_2010_binding-Mcl-1 | spearman | 0.133        | -0.054    | –       | –       |
| pg_CATR_CHLRE_Tsuboyama_2023_2AMI       | spearman | 0.066        | -0.086    | -0.039  | 0.135   |
| pg_CSN4_MOUSE_Tsuboyama_2023_1UFM       | spearman | -0.034       | 0.171     | 0.101   | -0.088  |
| pg_F7YBW8_MESOW_Ding_2023               | spearman | -0.112       | 0.416     | –       | –       |



**Figure 4: Controlled mechanism ablations.** (a) Noul candidate phrasing on three seed-averaged stable checkpoints (same weights; only candidate text varies). No phrasing reaches the per-task-head reference (0.629, dashed); the best (“false: no / true: yes”,  $0.553 \pm 0.023$ ) beats the *trained* phrasing (hatched) on all three seeds — reversing the single-checkpoint finding, i.e., inference-level ablations are also checkpoint-dependent. (b) Frozen VLM on synthetic property maps (multimodal route-A pilot): plain candidate likelihood collapses to one class (acc 0.040, 1/7 classes used); contrastive decoding removes the collapse (0.140, 4/7) but remains at chance — decoding fixed, signal absent.

behavior (4/7 classes, accuracy 0.140) but sits at chance ( $1/7 = 0.143$ ), far below the majority baseline (0.325) and the text-only frozen LM (0.320). Image sanity checks rule out rendering failure (per-image statistics and cross-image correlations normal). Conclusion: route A (synthetic renders) with a frozen VLM is a dead end; the reusable finding is that *frozen-VLM candidate-likelihood evaluation should default to contrastive decoding*. Real-image localization (HPA) and fine-tuned VLM arms are a separate work.

## D Data, licensing, and disclosures

**Data and licensing.** All sources are public datasets with license, revision and SHA-256 recorded in download receipts; no human-subject data. RNAcompete is the 14-RBP test subset (not the full  $\sim 200$ ); DeepGOZero/PEER/DeepSEA remain in the catalog as needs-upstream-work and are not claimed as coverage. The splice label-convention doubt (Section 3.3) is explicitly recorded for independent verification.

**LLM-use disclosure.** [To be finalized per venue checklist: writing assistance disclosure.]

**Table 7:** Noul AUROC on GB tasks for six candidate phrasings, evaluated on three seed-averaged stable checkpoints (same weights per column; only rendered candidate text varies). No phrasing reaches the per-task-head reference (0.629). The trained phrasing is marked †.

| Phrasing                     | s1    | s2    | s3    | mean±std    |
|------------------------------|-------|-------|-------|-------------|
| false: no / true: yes        | 0.572 | 0.568 | 0.521 | 0.553±0.023 |
| false / true                 | 0.566 | 0.523 | 0.541 | 0.543±0.018 |
| proposition (“yes, it does”) | 0.517 | 0.513 | 0.549 | 0.526±0.016 |
| no / yes (trained)†          | 0.477 | 0.536 | 0.510 | 0.508±0.024 |
| descriptive                  | 0.527 | 0.506 | 0.431 | 0.488±0.041 |
| neg / pos words              | 0.513 | 0.468 | 0.472 | 0.484±0.020 |

**Table 8:** Cross-family score transfer (TAPE held out; DeepSTARR and ProteinGym score families in training). Spearman per seed; “abs” = trained absolute value. Stability’s positive  $\Delta$  reflects a pathological frozen baseline (−0.218) regressing toward zero, not genuine transfer.

| Task              |     | frozen | tr. s1 | tr. s2 | tr. s3 | $\Delta$ mean±std |
|-------------------|-----|--------|--------|--------|--------|-------------------|
| tape_fluorescence | abs | 0.037  | -0.008 | 0.026  | -0.075 | -0.056±0.042      |
| tape_stability    | abs | -0.218 | -0.018 | 0.020  | -0.179 | +0.159±0.086      |

## E Reproducibility

Every figure and table regenerates from on-disk JSONs via one script per artifact (`paper_v2/figures/gen_*.py`); no metric is hardcoded in the plotting or table code. Each run persists `run_config.json` (flags, seeds, caps), `train_trace.jsonl` (per-step loss/grad) and `dev_curve.jsonl` (per-family dev scores); evaluation subsets are drawn with a fixed sampling seed (20261001) rather than first- $N$  — a bug we found and fixed: class-ordered test files plus first- $N$  sampling produced a spurious noul AUROC of 0.387 that looked like a polarity bug but was a sampling artifact. Hardware: single H20/A800-class GPU per run; nondeterminism is absorbed by the seed protocol of Section 4.3.